Artificial Intelligence - A Key Enabler for the Next-Generation Solutions

HITACHI
Inspire the Next



Among the many technologies that are shaping India's future, one stands tall and that is Artificial Intelligence. India's *National Strategy for Artificial Intelligence,* crafted by NITI Aayog, unveils the boundless potential of Artificial Intelligence in ensuring economic growth, good governance, and sustainable society.

At the forefront of responsible innovation, Hitachi India R&D Centre paves the way for social progress using Artificial Intelligence. Hitachi has recently established "Generative AI Centre" which plays a decisive role in accelerating value creation and promoting safe, effective and ethical use of Artificial Intelligence.

On business front, a key thrust of Hitachi India R&D is research on the means and mechanisms for enabling a section of the society that is still not included digitally or financially. For example, Hitachi India R&D works on *voice authentication and automated speech recognition for Indian vernaculars,* to facilitate inclusive banking and digital payments. Driven by data and advanced analytics, R&D has developed a system of convenience for retail merchants who are key partners in Hitachi's payment ecosystem.

On technology implementation, a key challenge is to tap the appropriate place for placing the automated decision-making engine in the channel of human-computer interaction. For example, we need to balance between "smart edges" (i.e., Hitachi devices with peripheral intelligence) and "control-freak central platform" (akin to central nervous system). Both have their advantages and disadvantages, but for a solution to be sustainable we need to distribute the automated intelligence appropriately and center around humans continuously.

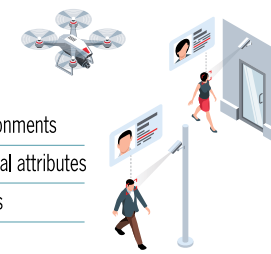
> As we embrace the future, it's not about AI replacing human intelligence but rather augmenting it for better living.
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> **Kingshuk Banerjee**
> Head of R&D, Hitachi India Pvt. Ltd.

Video Analytics for a Safer Society

In the pursuit of a safer and more secure society, law enforcement agencies in India are embracing modern surveillance systems paired with video analytics. Powered by Hitachi R&D's resilient technological prowess and the versatile Hitachi Visualization Suite, the team is crafting a unique, AI-driven solution tailored for India.

R&D's Custom Algorithms & Resilient Crime Prevention Solutions

- Accurately detect objects in challenging environments
- Identify individuals based on attire and personal attributes
- Condense hours of video footage into minutes
- Analyze uncommon occurrences



Voice-Based Solutions for Empowering the Underprivileged

In a country with a significant rural-urban divide and varying literacy levels, voice-based AI interaction is poised to be a revolution in India's inclusive growth odyssey. Leading this progressive change, Hitachi India R&D has pioneered an Automatic Speech Recognition (ASR) platform for Indian vernacular languages, powered by cutting-edge AI and Machine Learning. This platform enables seamless human-machine interaction through voice commands, bridging the literacy gap and improving overall quality of life.

Automatic Speech Recognition Platform Capabilities

- Support for diverse languages, dialects & accents
- Scalability across India
- Intuitive interface
- Hitachi's proprietary deep signal processing & noise reduction capabilities
- 24/7 availability





ATMs



Point of Sale (POS)

Merchant-Centric Solutions to Drive Digital Penetration

With India emerging as a leading digital economy and record-breaking digital transactions, Hitachi India R&D is offering solutions to drive digital penetration within the banking and payments industry. Strengthening India's digital payment ecosystem, R&D is providing merchant-centric analytics solutions to a prominent public sector bank.

Hitachi India R&D is committed to optimizing the payments ecosystem through rigorous research and data analytics, with a specific focus on enhancing operational efficiency across various payment channels, including ATMs and Point-of-Sale (PoS).

Conclusion

Social Innovation, Data Democratization and Digital-Public-Good are the cornerstones of governance in modern India. Hitachi India R&D, with its data-driven approach is a fully subscribed party to the Digital India initiative. It attempts to leverage the tremendous power of data to design and develop the next-generation solutions for a society that is Connected, Informed and Insightful.

Scan for Social Innovation Business Site